LOOP: AI-POWERED VIRTUAL FITTING

Loop is an innovative, computer vision-driven virtual fitting platform designed to **eliminate apparel sizing friction**, starting with the historically complex, **$3.2B annual sizing and return friction problem** in the women's bra industry. By replacing invasive traditional measurements with a quick, privacy-first 3D scan, Loop matches consumers to their perfect size and style across various brands with over **92% fit accuracy**.

CRACKING THE CODE TO SIZE CHARTS

And why we're starting with the hardest formula first.

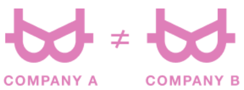
Size charts aren't going anywhere, but they are notoriously hard for consumers to navigate. Every brand uses proprietary math and sizing logic, forcing online shoppers to guess their way through confusing, flat grids that rarely reflect real-world body contours.

We aren't throwing out the size chart; **we're decoding it**. By capturing an individual's exact 3D physical contours, Loop acts as a **smart vector translator**, mapping unique 3D body geometry to any brand's specific sizing rules to deliver a deterministic fit match on the first try.

To prove our computer vision pipeline can untangle any size chart in retail, we launched by tackling the **most structurally sensitive and notoriously inconsistent sizing grid** in fashion: women's intimates.

Why Bras?

If we can accurately translate body geometry to a bra size chart, we can scale to any garment category in e-commerce.

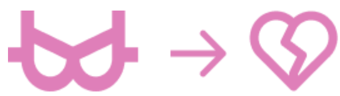


Company A ≠ Company B

Brands use completely different sizing math, even between their own styles.

50% Return Rates

Online bra retailers face massive return rates due to fit uncertainty.

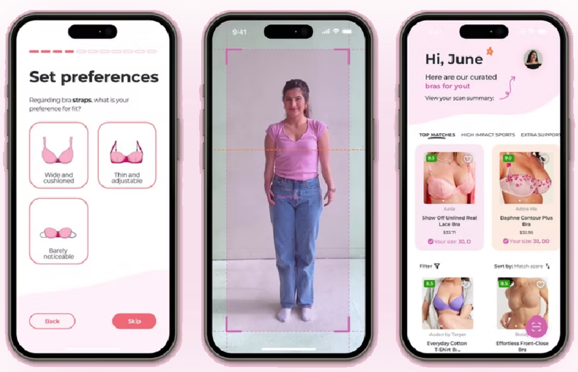


Health & Confidence

Ill-fitting bras lead to back pain, nerve pressure, and loss of confidence.

ENTER LOOP: THE SMART FIT TRANSLATOR IN YOUR POCKET

We built Loop to bridge the gap between real human bodies and online apparel sizing, replacing error-prone self-conducted tape measurements with instant mobile computer vision.



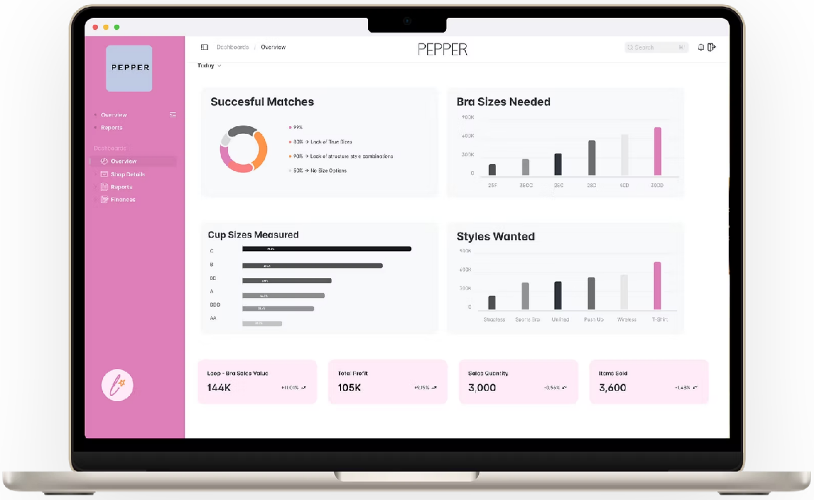
B2B PLATFORM & TECHNICAL ARCHITECTURE

RETAILER INSIGHTS & TECHNICAL PIPELINE

RETAILER INSIGHTS: ENTERPRISE TELEMETRY & OBSERVABILITY

Real-time sizing telemetry, fit anomaly detection & supply chain demand intelligence.

Bra companies partnering with Loop leverage our enterprise telemetry and observability platform to gain precise consumer measurement data beyond band and bust size. By ingesting real-time fit telemetry and sizing error anomalies, merchants track size demand trends, style match friction, sales conversion impact, and real-world consumer preferences—helping brands optimize manufacturing, inventory allocation, and supply chain logistics.



HOW IT WORKS: TECHNICAL COMPUTER VISION PIPELINE

Client-side mobile computer vision translating raw body math into brand size recommendations.

1

3D Mobile Scan

ON-DEVICE MEDIAPIPE

On-device MediaPipe computer vision running locally on standard smartphones (accelerated via CoreML & Metal APIs) mapping 3D surface geometry, volume distribution, and body contours.

2

SMPL 3D Mesh Engine

BEHIND THE SCENES

Landmark coordinate vectors fit a parametric 3D body shape model (SMPL) calibrated to height baselines, translating body math into cup & band sizes.

3

Validated Fit

92% FIT ACCURACY

Delivers brand-by-brand size recommendations benchmarked against master tailor ground-truth tape baselines (MAE 0.3" landmark variance).

SECURITY, VALIDATION & MARKET OPPORTUNITY

PRIVATE BY DESIGN

Your body is your business. We keep it that way.

- **Zero Photo or Video Server Ingress:** Raw RGB camera frames are processed locally on-device and discarded immediately in volatile memory. Zero photos or videos ever touch a server.
- **Client-Side Landmark Mesh Extraction:** Real-time feature extraction strips video frames instantly on-device, converting raw pixels into an **anonymous 3D mathematical SMPL landmark mesh**.
- **Encrypted Vector Storage:** User profiles consist strictly of encrypted geometric coordinate vectors (β shape parameters) and volume distributions, never identifiable visual images.

REAL-WORLD VALIDATION, EVALS & MOMENTUM

Proven technical execution, primary field research & academic recognition.

- **Model Evaluation & Evals Framework:** Benchmark accuracy was measured directly against professional master tailor physical tape measurements as our near-100% ground-truth baseline, achieving **92% fit accuracy** (MAE = 0.3" landmark variance).
- **0-to-1 Technical Engineering Leadership:** Managed 6 software engineers, ML developers, and UI/UX designers across agile 2-week sprints from initial architecture to live production pilots.
- **Deep Primary Field Research:** Built on direct qualitative and quantitative insights from **150+ user interviews and 35 bra brand merchant interviews**.
- **Award-Winning Innovation & Funding:** Secured **$15K non-dilutive grant capital** | 🏆 Semifinalist, Wharton Startup Challenge (UPenn) | 🏆 Winner, New Venture Seed Competition (USC) | 🏆 Winner, Pitch Prize Competition (USC IYA)

THE BIG PICTURE: BEYOND THE BRA

Scaling from intimates to universal apparel fit translation across e-commerce.

While we launched with the hardest sizing grids first to **establish technical credibility and user trust**, our vision expands across all apparel categories in e-commerce. Once Loop captures a user's unique 3D body geometry, their personal fit profile becomes a **universal fit translator** for the rest of retail.